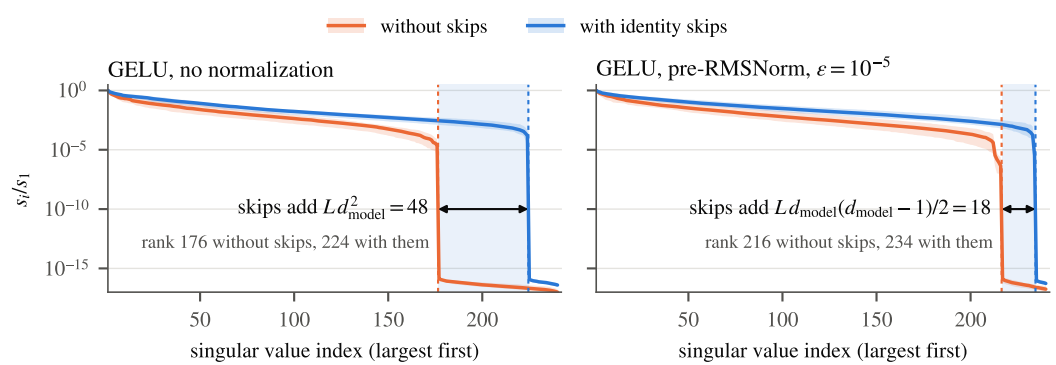
without skips
with identity skips
GELU, no normalization
GELU, pre-RMSNorm, $\varepsilon = 10^{-5}$
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
skips add $Ld_{\text{model}}^2 = 48$
rank 176 without skips, 224 with them
skips add $Ld_{\text{model}}(d_{\text{model}} - 1)/2 = 18$
rank 216 without skips, 234 with them
50
100
150
200
singular value index (largest first)
singular value index (largest first)